

Johnathan W. Carroll

Infrastructure, Security & AI Platform Engineering Leader

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Hands-on technology leader with 25+ years across government, enterprise IT, ISP operations, entrepreneurship, and applied AI. Combines systems architecture and operational leadership with infrastructure modernization, cybersecurity, disaster recovery, and local AI platform engineering.

Current Experience

Pascua Yaqui Tribe Government

Tucson, Arizona

IT Infrastructure Manager
Senior System Administrator
Security Analyst

Aug 2023 - Present
Jan 2021 - Aug 2023
Jan 2019 - Jan 2021

- Lead a nine-person Infrastructure/SecOps organization supporting approximately 1,500 employees across seven sites and critical government services.
- Serve as sole systems infrastructure engineer and systems/security architect, reporting to the IT Director; own compute, virtualization, storage, capacity, standards, and lifecycle planning.
- Consolidated legacy platforms around Nutanix AHV, Cisco UCS, and Pure Storage, producing approximately \$250,000 in annual savings and a 50% infrastructure footprint reduction.
- Lead vendor evaluation, RFPs, licensing, procurement recommendations, and infrastructure CapEx/OpEx planning; led Darktrace adoption and layered security modernization.
- Designed Veeam-based backup/DR architecture with immutable backups and geographically separated recovery; originated an estimated \$6M data-center/SOC/NOC program, still in design/procurement and partially funded.
- Inform the emerging Tribal AI architecture, security, governance, and policy program through independent research and Attorney General collaboration; development and scoped pilots remain distinct from production.

TechSoft Systems

2012 - Present

Founder, Owner & Operator

- Founded a technology consultancy, now primarily independent applied AI R&D alongside the full-time Tribal role; build local AI labs and GPU/server systems.
- Engineered a dual RTX PRO 6000 Blackwell platform with 192 GB aggregate GPU memory and 512 GB ECC DDR5; convert and optimize base models to NVFP4 with NVIDIA Model Optimizer for vLLM deployment.
- Work across CUDA, tensor parallelism, FP8 KV caching, speculative decoding, model residency, embeddings/reranking, agent memory, observability, and reproducible AI-assisted engineering.

Selected AI Research

Independent applied R&D and AI-assisted engineering; not commercial or completed Tribal production deployments.

NVIDIA AI-Q Research Assistant

Built a self-hosted AI-Q/NeMo platform and reproducible evidence evaluation; repaired a specific citation-accounting defect, validated with 196 focused passing tests.

PIPER Intent Gateway

Architected a Hermes advisory intent/memory plugin with deterministic contracts, monitoring, human review, historical regression coverage, and a selected 106-decision audit.

Ogden / Memory Workbench

Evaluated 280 conclusions across 72 cases after retries/recovery, yielding 19 human-review proposals without mutating live memory.

Previous Experience

Tucson Electric Power 2017 - 2019
Security Analyst - Contract

- Led CCTV/access-control modernization and integration into IT across headquarters, power generation, and Arizona substations; security scope within a \$40M+ program, including its \$30M data-center conversion.

TBM Equities LLC / Sahara Apartments Jul 2013 - Jul 2018
IT Director / Senior System Administrator

- Restored ransomware-affected services and business operations within 24 hours through an initial TechSoft engagement. Later reduced hardware footprint 60% and energy costs approximately \$8K/year; delivered a 15-minute RPO for critical systems.

BestComm Networks 2006 - 2012
Technical Engineer / Supervisor

- Provided advanced Tier 3 engineering and 24/7 NOC support within an international Best Western provider environment of approximately 3,400 properties; supervised staff and supported network modernization, monitoring, and complex escalations.

TheRiver ISP 2004 - 2006
Technical Manager

- Managed a 52-person technical operations organization serving approximately 26,000 subscribers; led staffing, escalations, carrier relationships, capacity planning, modernization, and the technical/workforce transition following acquisition.

Microsoft Support Contracts Approximately 2000 - 2004
Advanced Technical Support & Escalation

- Delivered Microsoft-authorized third-party support through Keen and later Convergys, with Microsoft-provided training: final-tier server case ownership in the earlier assignment and later Tier 2/3 Windows escalation and mentoring.

OneSmack.com LLC Mid-1990s - 2000 | Norfolk, Virginia
Founder, Owner & Operator

- Founded a web-development and technology business serving approximately 26 clients; delivered database-backed applications, onboarding/quoting workflows, hosting administration, custom systems, and office IT integration.

Technical Domains

Infrastructure	Nutanix AHV, VMware, Hyper-V, Cisco UCS, Pure Storage, NetApp, Veeam, Linux
Networking	Cisco Nexus/Catalyst, Palo Alto, VLAN/VXLAN, VPN
Security	Darktrace, Defender, Sentinel, Rapid7, Syxsense, identity, incident response
AI platforms	CUDA, Model Optimizer, NVFP4, vLLM, AI-Q/NeMo, Hermes, Honcho, MCP
Engineering	Docker, Python, Bash, PowerShell, SQL, TypeScript/JavaScript, FastAPI, GitHub Actions
Observability	Prometheus, Grafana, Loki, NVIDIA DCGM Exporter

Education & Affiliations

- Computer Programming & Digital Repair Certificates - Norfolk Technical Vocational Center
- Coursework - Tidewater Community College
- InfraGard Arizona Member; open-source contributor and advocate